Homework 8

Solution

1. Determine whether each of the following series converges absolutely, converges conditionally or diverges:

(a)

$$\sum_{n=1}^{\infty} \frac{n \cdot \sin\left(n^2\right)}{2^n + n}$$

Solution:

It holds that

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{n \cdot \left|\sin\left(n^2\right)\right|}{2^n + n} \le \sum_{n=1}^{\infty} \frac{n}{2^n + n} \le \sum_{n=1}^{\infty} \frac{n}{2^n}$$

Applying the root test, we get

$$L = \lim_{n\to\infty} \sqrt[n]{\frac{n}{2^n}} = \lim_{n\to\infty} \frac{\sqrt[n]{n}}{2} = \frac{1}{2} < 1$$

We conclude that $\sum_{n=1}^{\infty} \frac{n}{2^n} < \infty$ by the root test, and therefore $\sum_{n=1}^{\infty} |a_n| < \infty$ by the comparison test.

Thus, the series $\sum_{n=1}^{\infty} a_n$ converges absolutely.

converges absolutely	converges conditionally	diverges
yes	no	no

(b)

$$\sum_{n=1}^{\infty}(-1)^{n+1}\sqrt{\frac{\sqrt{n+1}-\sqrt{n}}{n+1}}$$

Solution:

First we check if the series converges absolutely

$$\sum_{n=1}^{\infty}|a_n| = \sum_{n=1}^{\infty}\sqrt{\frac{\sqrt{n+1}-\sqrt{n}}{n+1}} = \sum_{n=1}^{\infty}\sqrt{\frac{1}{(n+1)\left(\sqrt{n+1}+\sqrt{n}\right)}}$$

We choose $b_n = \frac{1}{n^{\frac{3}{4}}}$. We use the limit comparison test

$$L = \lim_{n\to\infty}\frac{a_n}{b_n} = \lim_{n\to\infty}\frac{\sqrt{\frac{1}{(n+1)\left(\sqrt{n+1}+\sqrt{n}\right)}}}{\left[\frac{1}{n^{\frac{3}{4}}}\right]} = \lim_{n\to\infty}\sqrt{\frac{n^{\frac{3}{2}}}{(n+1)\left(\sqrt{n+1}+\sqrt{n}\right)}} = \lim_{n\to\infty}\sqrt{\frac{1}{\left(1+\frac{1}{n}\right)\left(\sqrt{1+\frac{1}{n}}+1\right)}} = \frac{1}{\sqrt{2}}$$

We conclude that the series $\sum_{n=1}^{\infty}|a_n|$ diverges together with the $\frac{3}{4}$-Harmonic series, so the series $\sum_{n=1}^{\infty}a_n$ does not converge absolutely.

We now check if the series $\sum_{n=1}^{\infty}a_n$ is a Leibniz series.

Consider the sequence $(b_n)_{n=1}^{\infty}$ defined by $b_n = \sqrt{\frac{\sqrt{n+1}-\sqrt{n}}{n+1}}$ for every $n \in \mathbb{N}$. Note that $\sum_{n=1}^{\infty}a_n = \sum_{n=1}^{\infty}(-1)^{n+1}b_n$. We note that

$$\lim_{n\to\infty}b_n = \lim_{n\to\infty}\sqrt{\frac{\sqrt{n+1}-\sqrt{n}}{n+1}} = \lim_{n\to\infty}\frac{1}{n^{\frac{3}{4}}\sqrt{\left(1+\frac{1}{n}\right)\left(\sqrt{1+\frac{1}{n}}+1\right)}} = 0$$

Note also that $(b_n)_{n=1}^{\infty}$ is decreasing, as the denominator is positive and increasing. Thus, $\sum_{n=1}^{\infty}a_n$ is a Leibniz series and therefore converges.

converges absolutely	converges conditionally	diverges
no	yes	no

2. Let $(a_n)_{n=1}^{\infty}, (b_n)_{n=1}^{\infty}$ be two sequences of real numbers.

(a) Prove that if the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge absolutely, then the series $\sum_{n=1}^{\infty} (a_n - b_n)$ converges absolutely.

Solution:

It holds that

$$\sum_{n=1}^{\infty} |a_n - b_n| \overset{\Delta}{\leq} \sum_{n=1}^{\infty} (|a_n| + |-b_n|) \overset{\text{AOS}}{=} \sum_{n=1}^{\infty} |a_n| + \sum_{n=1}^{\infty} |b_n| < \infty$$

Thus, by the comparison test, $\sum_{n=1}^{\infty} |a_n - b_n| < \infty$ and therefore the series $\sum_{n=1}^{\infty} (a_n - b_n)$ converges absolutely.

(b) Give an example where the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge conditionally, and the series $\sum_{n=1}^{\infty} (a_n - b_n)$ converges conditionally.

Solution:

We choose $a_n = \frac{(-1)^{n+1}}{n}$ and $b_n = \frac{(-1)^n}{n}$ for every $n \in \mathbb{N}$.

We saw that the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge conditionally.

Note that

$$\sum_{n=1}^{\infty} (a_n - b_n) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} - (-1)^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} + (-1)^{n+1}}{n} = 2\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

thus the series $\sum_{n=1}^{\infty} (a_n - b_n)$ converges conditionally.

(c) Prove that if the series $\sum_{n=1}^{\infty} a_n$ converges conditionally and the series $\sum_{n=1}^{\infty} b_n$ converges absolutely, then the series $\sum_{n=1}^{\infty} (a_n + b_n)$ converges conditionally.

Solution:

ATC that the series $\sum_{n=1}^{\infty} (a_n + b_n)$ does not converge conditionally. First, as it is a sum of convergent series, then it converges by AOS. As it doesn't converge conditionally, it must therefore converge absolutely. By item (a), the series $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (a_n + b_n - b_n)$ converges absolutely. Contradiction.

(d) Conclude that the series $\sum_{n=1}^{\infty} \frac{\cos(n)+(-1)^{n+1}\cdot(n+1)}{(n+1)\ln^2(n+1)}$ converges conditionally.

Solution:

Consider the sequences $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ defined by $a_n = \frac{(-1)^{n+1}}{\ln^2(n+1)}$ and $b_n = \frac{\cos(n)}{(n+1)\ln^2(n+1)}$ for every $n \in \mathbb{N}$.

We note that

$$\lim_{n\to\infty} \frac{\ln^2(n+1)}{n} \underset{\text{Heine's lemma}}{=} \lim_{x\to\infty} \frac{\ln^2(x+1)}{x} \underset{L}{=} \lim_{x\to\infty} \frac{2\ln(x+1)\cdot\frac{1}{x+1}}{1} = \lim_{x\to\infty} \frac{2\ln(x+1)}{x+1} \underset{L}{=} \lim_{x\to\infty} \frac{2\cdot\frac{1}{x+1}}{1} = 0$$

Thus there exists a $k \in \mathbb{N}$ such that for every $n \geq k$ we have $\ln^2(n+1) \leq n$.

Thus

$$\sum_{n=k}^{\infty} \frac{1}{\ln^2(n+1)} \geq \sum_{n=k}^{\infty} \frac{1}{n} = \infty$$

Therefore the series $\sum_{n=k}^{\infty} \frac{1}{\ln^2(n+1)}$ diverges and therefore the series $\sum_{n=1}^{\infty} \frac{1}{\ln^2(n+1)}$ diverges (has a divergent tail).

Therefore the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\ln^2(n+1)}$ does not converge absolutely.

Clearly, the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\ln^2(n+1)}$ is a Leibniz series, thus the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\ln^2(n+1)}$ converges conditionally.

As

$$\sum_{n=1}^{\infty} |b_n| = \sum_{n=1}^{\infty} \left| \frac{\cos(n)}{(n+1)\ln^2(n+1)} \right| \leq \sum_{n=1}^{\infty} \frac{1}{(n+1)\ln^2(n+1)} = \sum_{n=2}^{\infty} \frac{1}{n\ln^2(n)} < \infty$$

then the series $\sum_{n=1}^{\infty} b_n$ converges absolutely.

By item (c), we conclude that the series

$$\sum_{n=1}^{\infty} \frac{\cos(n)+(-1)^{n+1}\cdot(n+1)}{(n+1)\ln^2(n+1)} = \sum_{n=1}^{\infty} \left(\frac{\cos(n)}{(n+1)\ln^2(n+1)} + \frac{(-1)^{n+1}}{\ln^2(n+1)} \right) = \sum_{n=1}^{\infty} (a_n + b_n)$$

converges conditionally.

3. (a) Student A proved that the alternating Harmonic series is convergent in the following way:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \sum_{n=1}^{\infty} \left(\frac{1}{2n-1} - \frac{1}{2n} \right) = \sum_{n=1}^{\infty} \frac{1}{2n(2n-1)} \leq \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

Explain student A's mistake.

Solution:

The equality

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \sum_{n=1}^{\infty} \left(\frac{1}{2n-1} - \frac{1}{2n} \right)$$

already assumes that the alternating Harmonic series is convergent, as is required by the parentheses test.

It follows that his proof is circular.

(b) Student A proved that the alternating Harmonic series converges to zero in the following way:

As the series is a Leibniz series, it is convergent. Thus, there exists an $s \in \mathbb{R}$ such that

$$s = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

Thus,

$$\begin{aligned}
s = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots &= 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \dots \\
&= \left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) - \frac{1}{12} + \dots \\
&= \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \frac{1}{14} - \frac{1}{16} + \dots \\
&= \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \dots\right) = \frac{1}{2}s
\end{aligned}$$

Thus, $s = \frac{1}{2}s$ and therefore $s = 0$.

Explain student A's mistake.

Solution:

Student A make a mistake when changing the order of the summation of the series.

4. Prove or disprove each of the following statements:

(a) The series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\ln n}{\sqrt{n}+1}$ converges absolutely.

Solution:

False. Note that

$$\sum_{n=1}^{\infty}\left|(-1)^{n+1}\frac{\ln n}{\sqrt{n}+1}\right| = \sum_{n=1}^{\infty}\frac{\ln n}{\sqrt{n}+1} \geq \sum_{n=3}^{\infty}\frac{\ln n}{\sqrt{n}+1} \geq \sum_{n=3}^{\infty}\frac{1}{\sqrt{n}+1} \geq \sum_{n=3}^{\infty}\frac{1}{2\sqrt{n}} = \frac{1}{2}\sum_{n=3}^{\infty}\frac{1}{\sqrt{n}} \overset{\text{tail of }\frac{1}{2}\text{-harmonic}}{=} \infty$$

Thus the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\ln n}{\sqrt{n}+1}$ does not converge absolutely.

(b) The series $1-\frac{1}{2}+\frac{2}{3}-\frac{1}{3}+\frac{2}{4}-\frac{1}{4}+\frac{2}{5}-\frac{1}{5}+\ldots$ is divergent.

Solution:

True. ATC that the series is convergent. Then there exists an $s\in\mathbb{R}$ such that $s=1-\frac{1}{2}+\frac{2}{3}-\frac{1}{3}+\frac{2}{4}-\frac{1}{4}+\frac{2}{5}-\frac{1}{5}+\ldots$ and thus, by the parentheses test,

$$s=\left(1-\frac{1}{2}\right)+\left(\frac{2}{3}-\frac{1}{3}\right)+\left(\frac{2}{4}-\frac{1}{4}\right)+\left(\frac{2}{5}-\frac{1}{5}\right)+\cdots=\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\cdots=\sum_{n=2}^{\infty}\frac{1}{n}\overset{\text{tail of Harmonic}}{=}\infty$$

Contradiction.

(c) Let $(a_n)_{n=1}^{\infty}, (b_n)_{n=1}^{\infty}$ be two sequences of real numbers.

Suppose that $0\leq a_n\leq b_n$ for every $n\in\mathbb{N}$, $\lim\limits_{n\to\infty}b_n=0$ and $(b_n)_{n=1}^{\infty}$ is decreasing.

Then the series $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ is convergent.

Solution:

False. We choose $b_n=\frac{1}{n}$ and $a_n=\begin{cases}0 & n=2k\\ \frac{1}{n} & n=2k-1\end{cases}$ for every $n\in\mathbb{N}$.

Clearly $0\leq a_n\leq b_n$ for every $n\in\mathbb{N}$, $\lim\limits_{n\to\infty}b_n=0$ and $(b_n)_{n=1}^{\infty}$ is decreasing.

But

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n=\sum_{k=1}^{\infty}\frac{(-1)^{2k}}{2k-1}=\sum_{k=1}^{\infty}\frac{1}{2k-1}\geq\sum_{k=1}^{\infty}\frac{1}{2k}=\frac{1}{2}\sum_{k=1}^{\infty}\frac{1}{k}=\infty$$

and therefore $\sum_{k=1}^{\infty}\frac{1}{2k-1}=\infty$ and therefore $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ is divergent.

(d) The series $\sum_{n=1}^{\infty}\left(\sqrt{n^3+(-1)^{n+1}}-n^{1.5}\right)$ is convergent.

Solution:

We prove that the series converges absolutely by using the limit comparison test. Choose $b_n = \boxed{\frac{1}{n^{1.5}}}$ for every $n \in \mathbb{N}$. Then

$$L = \lim_{n\to\infty}\frac{a_n}{b_n} = \lim_{n\to\infty}\frac{\left|\sqrt{n^3+(-1)^{n+1}}-n^{1.5}\right|}{\left[\frac{1}{n^{1.5}}\right]} = \lim_{n\to\infty}\left[n^{1.5}\left|\sqrt{n^3+(-1)^{n+1}}-n^{1.5}\right|\right] \overset{\text{conjugate}}{=} \lim_{n\to\infty}\frac{n^{1.5}}{\sqrt{n^3+(-1)^{n+1}}+n^{1.5}}$$

$$= \lim_{n\to\infty}\frac{1}{\sqrt{1+\frac{(-1)^{n+1}}{n^3}}+1} = \frac{1}{2}$$

Thus, the series converges absolutely together with the 1.5-Harmonic series (and therefore the series converges).

(e) It holds that $\sum_{n=1}^{\infty}\frac{1}{n(2n+1)} = 2-\ln(4)$.

Note: you may assume that $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n} = \ln(2)$.

Solution:

True. First, by the comparison test

$$\sum_{n=1}^{\infty}\frac{1}{n(2n+1)} \le \sum_{n=1}^{\infty}\frac{1}{n^2} \overset{\frac{1}{2}-\text{Harmonic}}{<} \infty$$

and therefore $\sum_{n=1}^{\infty}\frac{1}{n(2n+1)} < \infty$. Thus, there exists an $s \in \mathbb{R}$ such that $s = \sum_{n=1}^{\infty}\frac{1}{n(2n+1)}$.

We now show that $s = 2-\ln(4)$.

Note that the series $\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\dots$ converges by the Leibniz test.

It follows that

$$\frac{1}{2}s = \sum_{n=1}^{\infty}\frac{1}{2n(2n+1)} = \sum_{n=1}^{\infty}\left[\frac{1}{2n}-\frac{1}{2n+1}\right] = \left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{4}-\frac{1}{5}\right)+\dots$$

$$\overset{\text{parenatheses test}}{=} \frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\dots$$

and therefore

$$1-\frac{1}{2}s = 1-\left(\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}\right) = 1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\dots = \ln(2)$$

It follows that $\frac{1}{2}s = 1-\ln(2)$ and therefore $s = 2-2\ln(2) = 2-\ln(4)$.

5. Find all values of $\alpha \in \mathbb{R}$ for which the series $\sum_{n=1}^{\infty} \left(n^3 \sin\left(\frac{1}{n}\right) - n^2 + \alpha\right)$ is convergent.

Solution:

On the one hand, if the series converges, then by the necessary criterion we have

$$\lim_{n\to\infty} \left(n^3 \sin\left(\frac{1}{n}\right) - n^2 + \alpha\right) = 0.$$

On the other hand

$$\lim_{n\to\infty} \left(n^3 \sin\left(\frac{1}{n}\right) - n^2 + \alpha\right) \overset{\text{Heine's lemma}}{=} \lim_{x\to\infty} \left(x^3 \sin\left(\frac{1}{x}\right) - x^2 + \alpha\right) \overset{t=\frac{1}{x}}{=} \lim_{t\to 0^+} \left(\frac{\sin t}{t^3} - \frac{1}{t^2} + \alpha\right)$$

$$= \lim_{t\to 0^+} \frac{\sin t - t + \alpha t^3}{t^3} \underset{L}{\overset{\frac{0}{0}}{=}} \lim_{t\to 0^+} \frac{\cos t - 1 + 3\alpha t^2}{3t^2} \underset{L}{\overset{\frac{0}{0}}{=}} \lim_{t\to 0^+} \frac{-\sin t + 6\alpha t}{6t} \underset{L}{\overset{\frac{0}{0}}{=}} \lim_{t\to 0^+} \frac{-\cos t + 6\alpha}{6} = \frac{-1+6\alpha}{6} = \alpha - \frac{1}{6}$$

Thus

$$\alpha - \frac{1}{6} = 0 \Rightarrow \alpha = \frac{1}{6}$$

Thus if the series $\sum_{n=1}^{\infty} \left(n^3 \sin\left(\frac{1}{n}\right) - n^2 + \alpha\right)$ converges, then $\alpha = \frac{1}{6}$.

Now we prove that if $\alpha = \frac{1}{6}$, then the series $\sum_{n=1}^{\infty} \left(n^3 \sin\left(\frac{1}{n}\right) - n^2 + \frac{1}{6}\right)$ converges.

We prove that the series $\sum_{n=1}^{\infty} \left(n^3 \sin\left(\frac{1}{n}\right) - n^2 + \frac{1}{6}\right)$ converges absolutely.

We compute

$$\lim_{n\to\infty} \frac{\left|n^3 \sin\left(\frac{1}{n}\right) - n^2 + \frac{1}{6}\right|}{\left(\frac{1}{n^2}\right)} = \lim_{n\to\infty} \left|n^5 \sin\left(\frac{1}{n}\right) - n^4 + \frac{n^2}{6}\right| \overset{\text{Heine's lemma}}{=} \lim_{x\to\infty} \left|x^5 \sin\left(\frac{1}{x}\right) - x^4 + \frac{x^2}{6}\right| \overset{t=\frac{1}{x}}{=} \lim_{t\to 0^+} \left|\frac{\sin t}{t^5} - \frac{1}{t^4} + \frac{1}{6t^2}\right|$$

We compute the limit without the absolute value

$$\lim_{t\to 0^+} \left(\frac{\sin t}{t^5} - \frac{1}{t^4} + \frac{1}{6t^2}\right) = \lim_{t\to 0^+} \frac{\sin t - t + \frac{1}{6}t^3}{t^5} \underset{L}{\overset{\frac{0}{0}}{=}} \lim_{t\to 0^+} \frac{\cos t - 1 + \frac{1}{2}t^2}{5t^4} \underset{L}{\overset{\frac{0}{0}}{=}} \lim_{t\to 0^+} \frac{-\sin t + t}{20t^3} \underset{L}{\overset{\frac{0}{0}}{=}} \lim_{t\to 0^+} \frac{-\cos t + 1}{60t^2} \underset{L}{\overset{\frac{0}{0}}{=}} \lim_{t\to 0^+} \frac{\sin t}{120t} = \frac{1}{120} > 0$$

Thus

$$\lim_{t\to 0^+} \left|\frac{\sin t}{t^5} - \frac{1}{t^4} + \frac{1}{6t^2}\right| = \left|\frac{1}{120}\right| = \frac{1}{120} > 0.$$

and therefore the series $\sum_{n=1}^{\infty} \left(n^3 \sin\left(\frac{1}{n}\right) - n^2 + \frac{1}{6}\right)$ converges absolutely together with the 2-Harmonic series and therefore converges.

We conclude that the series $\sum_{n=1}^{\infty} \left(n^3 \sin\left(\frac{1}{n}\right) - n^2 + \alpha\right)$ converges if and only if $\alpha = \frac{1}{6}$.

6. Let $(a_n)_{n=1}^{\infty}$ a sequence of real numbers. Suppose that there exists an $L \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} a_n = L$.

Consider the sequence $(b_n)_{n=1}^{\infty}$ that is defined by $b_n = (-1)^n a_n$ for every $n \in \mathbb{N}$.

(a) Prove that there exists an $N_1 \in \mathbb{N}$ such that $\forall n \geq N_1$ we have $b_n \leq 1 + |L|$.

Solution:

As $\lim\limits_{n\to\infty} a_n = L$, then there exists an $N_1 \in \mathbb{N}$ such that for every $n \geq N_1$ we have

$$|a_n - L| < \boxed{1} \tag{*}$$

Choose $N = N_1$ and let $n \geq N$. Then $(*)$ hold for that n and therefore

$$b_n \leq |b_n| = |(-1)^n a_n| = |a_n| \overset{\Delta}{\leq} |a_n - L| + |L| < 1 + |L|$$

(b) Prove that if $(b_n)_{n=1}^{\infty}$ is convergent, then $\lim\limits_{n\to\infty} b_n = 0$.

Solution:

As $(b_n)_{n=1}^{\infty}$ is convergent, then there exists an $L_1 \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} b_n = L_1$. We need to show $L_1 = 0$.

As $\lim\limits_{n\to\infty} b_n = L_1$, then by the inheritence theorem we also have

$$\lim_{n\to\infty} b_{2n} = L_1 \qquad \text{and} \qquad = \lim_{n\to\infty} b_{2n-1} = L_1.$$

Similarly, as $\lim\limits_{n\to\infty} a_n = L$, then

$$\lim_{n\to\infty} a_{2n} = L \qquad \text{and} \qquad = \lim_{n\to\infty} a_{2n-1} = L.$$

It follows that

$$L_1 = \lim_{n\to\infty} b_{2n} = \lim_{n\to\infty} a_{2n} = L = \lim_{n\to\infty} a_{2n-1} = \lim_{n\to\infty} (-b_{2n-1}) = -L_1$$

and therefore $L_1 = 0$.

(c) Suppose that $(b_n)_{n=1}^{\infty}$ is increasing. Prove that the series $\sum_{n=1}^{\infty} a_n$ is convergent.

Solution:

Consider the sequence $(-b_n)_{n=1}^{\infty}$. We prove that it is decreasing and converges to zero.

(i) Let $n \in \mathbb{N}$. As $(b_n)_{n=1}^{\infty}$ is increasing then $b_n \leq b_{n+1}$. Thus $-b_n \geq -b_{n+1}$.

(ii) First, by item (a), the sequence $(b_n)_{n=N_1}^{\infty}$ is bounded from above by $1+|L|$. Since $(b_n)_{n=N_1}^{\infty}$ is also increasing, then $(b_n)_{n=N_1}^{\infty}$ is convergent, and therefore $(b_n)_{n=1}^{\infty}$ is also convergent. By item (b),

$$\lim_{n\to\infty} (-b_n) = -\lim_{n\to\infty} b_n = 0$$

It follows that

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (-1)^n b_n = \sum_{n=1}^{\infty} (-1)^{n+1} (-b_n) = \text{Leibniz}$$

and therefore the series converges by the Leibniz test.